FINAL YEAR PROJECT

PROJECT PROPOSAL

BS Information Technology — 7th Semester

VEHICLE NUMBER PLATE RECOGNITION

STUDENT & SUPERVISOR DETAILS

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Class / Section	BS IT — 7th Semester, 4M
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Project Title	Vehicle Number Plate Recognition
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Submitted in partial fulfillment of the requirements for the BS Information Technology degree.

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PROJECT PROPOSAL

Vehicle Number Plate Recognition System

1. Introduction

Vehicle Number Plate Recognition (VNPR) is a computer-vision system designed to identify and read vehicle registration numbers from images or video. The proposed project will capture a vehicle image, locate its number plate, process the plate image, and extract the registration characters using Optical Character Recognition (OCR). The recognized number can then be displayed and stored digitally for later reference.

2. Problem Statement

Manual recording and checking of vehicle registration numbers can be slow, repetitive, and prone to human error. A simple automated solution is needed to assist with vehicle identification and record keeping in places such as parking areas, campus entrances, and controlled-access locations.

3. Aim and Objectives

The main aim is to develop a prototype that automatically detects a vehicle number plate and recognizes its text. The objectives are to:

- Capture or upload vehicle images and video frames.
- Detect and crop the number plate region.
- Improve plate readability using image preprocessing.
- Recognize plate characters using OCR and display the result.
- Save recognized plate numbers with date/time for basic record keeping.
- Evaluate performance using test images and note common recognition errors.

4. Proposed Methodology

The system will follow these stages: (1) image/video input, (2) vehicle plate detection, (3) image preprocessing such as resizing, grayscale conversion and noise reduction, (4) character recognition using an OCR engine, (5) result validation and display, and (6) optional storage in a small database. Suitable tools may include Python, OpenCV, and an OCR library such as Tesseract; final tools will depend on implementation and available data.

5. Expected Outcomes

The expected outcome is a working prototype that detects number plates in supported images and attempts to read their characters automatically. It should provide a basic interface for viewing results and, if implemented, saving records. Accuracy may vary with lighting, camera angle, plate condition, image quality, and local plate formats.

6. Scope and Limitations

The initial scope is a prototype for still images or recorded video, with optional real-time camera input if time and hardware permit. The system is intended as an academic demonstration and decision-support tool, not as a replacement for official verification. Performance will depend on the quality and variety of the test data.

7. Conclusion

This project will explore the practical use of computer vision and OCR to automate vehicle number plate reading. The prototype can demonstrate how automated recognition may reduce manual data entry and support efficient vehicle record management.